

Dheerajkumar Ramasubramanian

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EDUCATION

Bachelor of Technology in Computer Science and Engineering - Artificial Intelligence (CSE-AI)

Year of completion: Expected Jul 2027

Amrita Vishwa Vidyapeetham, Bengaluru

Department of Computer Science and Engineering

CGPA: 7.58 (Upto 5th Semester)

Higher Secondary Education (Class XII)

Year of completion: Mar 2023

The Amaatra Academy, Bengaluru

Percentage: 84.4

Secondary Education (Class X)

Year of completion: Mar 2021

St. Peters School, Bengaluru

Percentage: 88.4

CREATIVE ACTIVITY

Projects

Glaucoma Progression Scoring System

- Developed deep learning framework for automated glaucoma assessment via optic disc (OD) and optic cup (OC) segmentation from retinal fundus images.
- Benchmarked segmentation architectures; best with MobileNet-UNet using shape-aware Focal-EIoU loss and Monte-Carlo Dropout uncertainty estimation.
- Integrated ODFormer-inspired multi-scale context and attention for anatomical consistency.
- Trained/validated on REFUGE dataset; automated glaucoma scoring via cup-to-disc ratio with confidence-aware support.

Dark Data Decarbonization: Carbon-Aware Cloud Storage Optimization Framework

- Designed AI-driven framework to detect/optimize dark data using Tencent Cloud traces.
- Integrated ML classification with carbon-aware algorithms (PSO, ACO, ARAS) and scheduling.
- Deployed on Minikube Kubernetes for automated tiering/deletion, reducing energy/carbon costs.

Microbial Bioremediation of Polluted Water

- Built AI model to predict bioremediation potential using genomic/environmental data.
- Integrated NCBI databases for genes (e.g., PETase); constructed feature datasets.
- Trained ML classifiers for pollutant prediction and bioremediation recommendations.

SERVICE & PROFESSIONAL DEVELOPMENT

Professional Development

Technical Skills

Programming Languages: Python, Java, C, C++, MATLAB

AI/ML & Data: Machine Learning, Deep Learning, TensorFlow, Scikit-learn, NumPy, Pandas, OpenCV

Tools & Platforms: Git, GitHub, Docker, Kubernetes, Minikube, AWS, SQL, REST APIs, Jupyter, MongoDB, Wireshark

Languages

English (Fluent), Tamil (Fluent), Hindi (Proficient).

Certifications / Short Courses

Complete Machine Learning with R Studio for ML (2025), C for Everyone: Programming Fundamentals, AWS Academy Graduate - AWS Academy Cloud Foundations, Great Learning - Computer Vision Essentials.